Emergent Communication Research Team Plan

This document outlines the research direction, learning roadmap, team structure, workflow, and first milestones for the emergent communication research team.

1. Research Vision

Primary domain: “Emergent Communication in Artificial Intelligence”

Core interests:

• AI agents inventing their own communication/language

• Coordination between multiple AI agents

Long-term goals may include:

• Research papers

• Open-source projects

• Patents

• Competition submissions

• Startup ideas

• Deep learning and research exposure

2. Research Philosophy

The team is not approaching this as a normal predefined project.

The first goal is NOT immediate novelty or publication.

The first objective is to build strong understanding and research capability.

The first few months should focus on:

• Learning foundations

• Understanding papers

• Building experiments

• Observing agent behavior

• Maintaining research notes

• Understanding emergent communication systems

3. First Major Milestone

“Build a complete small multi-agent communication system from scratch and understand how communication emerges between agents.”

Initial Project Idea:

A simple 2-agent gridworld environment

Agent A knows the target location

Agent B does not know the target location

Agent A can send only limited symbols/messages

Agent B must reach the target using those messages

Observe whether communication protocols emerge naturally

4. Core Topics to Learn

- ➔ Reinforcement Learning (RL)
- ➔ Multi-Agent Reinforcement Learning (MARL)
- ➔ Emergent Communication
- ➔ PyTorch
- ➔ Gymnasium
- ➔ PettingZoo
- ➔ Sender-Receiver Games
- ➔ Communication Bottlenecks
- ➔ Discrete Messaging Systems
- ➔ Experiment Analysis and Documentation

5. Team Structure

All members should understand the complete basics and overall workflow of the project.

Task division is for efficiency and deeper focus, not isolated knowledge.

Vijay:

- Research direction
- Experiment planning
- Architecture and integration
- Paper analysis
- Documentation

Owais:

- Coding implementations
- RL environments
- Training experiments

Sreedevikaa:

- Literature review
- Paper summaries
- Metrics and visualization
- Experiment analysis

6. Remote Team Workflow

Since the team works remotely, proper structure is important.

Maintain:

• Shared GitHub repository

• Shared research notes (Can use Notion)

• Shared documentation

• Regular discussion

• Experiment logs

• Continuous updates

7. Week-wise Learning Roadmap

Week 1 — Reinforcement Learning Foundations

Agents, environments, rewards

Episodes, states, actions

Q-learning intuition

Exploration vs exploitation

Basic PyTorch fundamentals

Goal: Understand how RL agents learn

Week 2 — Multi-Agent Reinforcement Learning

Multi-agent environments

Cooperative vs competitive agents

Partial observability

Gymnasium and PettingZoo basics

Goal: Understand how multiple agents interact

Week 3 — Emergent Communication Basics

Sender-receiver games

Communication bottlenecks

Limited vocabulary systems

Symbol emergence

Read foundational papers

Goal: Understand communication emergence

Week 4 — Building the First Project

Build the 2-agent gridworld

Create communication channels

Environment setup

Initial training/testing

Goal: Working prototype

Week 5 — Training and Analysis

Train the agents

Observe communication patterns

Measure coordination success

Analyze emergent behavior

Goal: Complete first foundational emergent communication system

8. Potential Applications

- Autonomous multi-agent AI systems
- AI-to-AI communication frameworks
- Swarm intelligence systems
- Intelligent robotics coordination
- Decentralized AI ecosystems
- AI safety and controllable communication
- Collaborative problem-solving agents
- Smart traffic and logistics systems
- Distributed decision-making systems
- Future AGI and autonomous agent research

9. Important Terms

Build > discussing grand ideas

Keep experiments simple initially

Document everything

Failed experiments are valuable

Maintain consistency

10. Recommended Resources

Books

- Reinforcement Learning: An Introduction — Sutton & Barto
- Deep Reinforcement Learning Hands-On — Maxim Lapan
- Artificial Intelligence: A Modern Approach — Russell & Norvig

Frameworks & Libraries

- PyTorch
- Gymnasium
- PettingZoo
- Weights & Biases (experiment tracking)

Research Platforms

- arXiv
- Papers With Code
- Google Scholar

Foundational Papers

- Learning to Communicate with Deep Multi-Agent Reinforcement Learning
- Emergence of Grounded Compositional Language in Multi-Agent Populations
- Attention Is All You Need

Educational Channels

- Yannic Kilcher
- DeepMind
- Two Minute Papers

“This document and initial research structure were generated with the assistance of ChatGPT based on the team’s ideas, research direction, and objectives, and were further reviewed and edited manually.”